

Abstract

This thesis studies statistical inference for constant-stepsizes linear stochastic approximation with Markovian noise and Richardson–Romberg extrapolation. The goal of the work is to describe how confidence statements can be constructed for Polyak–Ruppert averaged iterates when the observations are generated by a dependent Markov chain and the algorithm is run with non-vanishing step sizes. The main tasks are to formulate the linear stochastic approximation model, separate the deterministic and stochastic components of the averaged error, analyze the effect of Richardson–Romberg coupling of the step sizes α and 2α , and connect the resulting distributional approximation with long-run variance estimation.

The methodology combines finite-sample decompositions of linear recursions, Lyapunov stability arguments, Poisson-equation tools for geometrically ergodic Markov chains, martingale approximation, Berry–Esseen type bounds, and overlapping batch-means variance estimation. The empirical part is based on simulations for representative linear stochastic approximation models. These experiments compare ordinary averaging with the Richardson–Romberg estimator and examine the interaction between bias reduction and covariance estimation.

The main result is a non-asymptotic normal approximation for scalar projections of the Richardson–Romberg averaged estimator, together with the identification of the covariance target $\Sigma_\infty = \bar{A}^{-1} \Sigma_\varepsilon^M \bar{A}^{-T}$. The analysis shows which terms are cancelled by extrapolation, which residual terms are controlled by mixing and stability, and why burn-in and variance estimation are essential for confidence intervals. The numerical results support the theoretical conclusion that Richardson–Romberg extrapolation can substantially reduce step-size bias while preserving the standard inference target when the covariance estimator is calibrated appropriately.